

Your Name:

ID #:

Worksheet: Inverse Trig Function Values

Note: To review inverse trig functions, go to the learning target “I”

1. Compute the following values

(a) $\sin^{-1}(\frac{1}{2})$

Solution:

$$\begin{aligned}\text{let } \theta &= \sin^{-1}(\frac{1}{2}), \text{ then} \\ \sin(\theta) &= \frac{1}{2} \text{ with } \theta \in [-\frac{\pi}{2}, \frac{\pi}{2}], \\ \implies \theta &= \frac{\pi}{6} \\ \implies \sin^{-1}(\frac{1}{2}) &= \frac{\pi}{6}\end{aligned}$$

(d) $\arcsin(-\frac{\sqrt{2}}{2})$

Solution:

$$\begin{aligned}\text{let } \theta &= \arcsin(-\frac{\sqrt{2}}{2}), \text{ then} \\ \sin(\theta) &= -\frac{\sqrt{2}}{2} \text{ with } \theta \in [-\frac{\pi}{2}, \frac{\pi}{2}] \\ \implies \theta &= -\frac{\pi}{4} \\ \implies \arcsin(-\frac{\sqrt{2}}{2}) &= \boxed{-\frac{\pi}{4}}\end{aligned}$$

(b) $\cos^{-1}(\frac{1}{2})$

Solution:

$$\begin{aligned}\text{let } \theta &= \cos^{-1}(\frac{1}{2}), \text{ then} \\ \cos(\theta) &= \frac{1}{2} \text{ with } \theta \in [0, \pi] \\ \implies \theta &= \frac{\pi}{3} \\ \implies \cos^{-1}(\frac{1}{2}) &= \boxed{\frac{\pi}{3}}\end{aligned}$$

(e) $\arccos(\frac{\sqrt{3}}{2})$

Solution:

$$\begin{aligned}\text{let } \theta &= \arccos(\frac{\sqrt{3}}{2}), \text{ then} \\ \cos(\theta) &= \frac{\sqrt{3}}{2} \text{ with } \theta \in [0, \pi] \\ \implies \theta &= \frac{\pi}{6} \\ \implies \arccos(\frac{\sqrt{3}}{2}) &= \boxed{\frac{\pi}{6}}\end{aligned}$$

(c) $\tan^{-1}(-1)$

Solution:

$$\begin{aligned}\text{let } \theta &= \tan^{-1}(-1), \text{ then} \\ \tan(\theta) &= -1 \text{ with } \theta \in (-\frac{\pi}{2}, \frac{\pi}{2}) \\ \implies \theta &= -\frac{\pi}{4} \\ \implies \tan^{-1}(-1) &= \boxed{-\frac{\pi}{4}}\end{aligned}$$

(f) $\tan^{-1}(-\frac{\sqrt{3}}{3})$

Solution:

$$\begin{aligned}\text{let } \theta &= \tan^{-1}(-\frac{\sqrt{3}}{3}), \text{ then} \\ \tan(\theta) &= -\frac{\sqrt{3}}{3} \text{ with } \theta \in (-\frac{\pi}{2}, \frac{\pi}{2}) \\ \implies \theta &= -\frac{\pi}{6} \\ \implies \tan^{-1}(-\frac{\sqrt{3}}{3}) &= \boxed{-\frac{\pi}{6}}\end{aligned}$$

(g) $\tan^{-1}(\sqrt{3})$

Solution:

$$\begin{aligned} \text{let } \theta &= \tan^{-1}(\sqrt{3}), \text{ then} \\ \tan(\theta) &= \sqrt{3} \text{ with } \theta \in (-\frac{\pi}{2}, \frac{\pi}{2}) \\ \implies \theta &= \frac{\pi}{3} \\ \implies \tan^{-1}(\sqrt{3}) &= \boxed{\frac{\pi}{3}} \end{aligned}$$

(k) $\arccos(-\frac{\sqrt{3}}{2})$

Solution:

$$\begin{aligned} \text{let } \theta &= \arccos(-\frac{\sqrt{3}}{2}), \text{ then} \\ \cos(\theta) &= -\frac{\sqrt{3}}{2} \text{ with } \theta \in [0, \pi] \\ \implies \theta &= \frac{5\pi}{6} \\ \implies \arccos(-\frac{\sqrt{3}}{2}) &= \boxed{\frac{5\pi}{6}} \end{aligned}$$

(h) $\tan^{-1}(-\sqrt{3})$

Solution:

$$\begin{aligned} \text{let } \theta &= \tan^{-1}(-\sqrt{3}), \text{ then} \\ \tan(\theta) &= -\sqrt{3} \text{ with } \theta \in (-\frac{\pi}{2}, \frac{\pi}{2}), \\ \implies \theta &= -\frac{\pi}{3} \\ \implies \tan(\theta) &= -\sqrt{3} = \boxed{-\frac{\pi}{3}} \end{aligned}$$

(l) $\cos^{-1}(\frac{\sqrt{2}}{2})$

Solution:

$$\begin{aligned} \text{let } \theta &= \cos^{-1}(\frac{\sqrt{2}}{2}), \text{ then} \\ \cos(\theta) &= \frac{\sqrt{2}}{2} \text{ with } \theta \in [0, \pi] \\ \implies \theta &= \frac{\pi}{4} \\ \implies \cos^{-1}(\frac{\sqrt{2}}{2}) &= \boxed{\frac{\pi}{4}} \end{aligned}$$

(i) $\sin^{-1}(1)$

Solution:

$$\begin{aligned} \text{let } \theta &= \sin^{-1}(1), \text{ then} \\ \sin(\theta) &= 1 \text{ with } \theta \in [-\frac{\pi}{2}, \frac{\pi}{2}] \\ \implies \theta &= \frac{\pi}{2} \\ \sin^{-1}(1) &= \boxed{\frac{\pi}{2}} \end{aligned}$$

(m) $\sin^{-1}(0)$

Solution:

$$\begin{aligned} \text{let } \theta &= \sin^{-1}(0), \text{ then} \\ \sin(\theta) &= 0 \text{ with } \theta \in [-\frac{\pi}{2}, \frac{\pi}{2}] \\ \implies \theta &= 0 \\ \implies \sin^{-1}(0) &= \boxed{0} \end{aligned}$$

(j) $\arcsin(\frac{\sqrt{3}}{2})$

Solution:

$$\begin{aligned} \text{let } \theta &= \arcsin(\frac{\sqrt{3}}{2}), \text{ then} \\ \sin(\theta) &= \frac{\sqrt{3}}{2} \text{ with } \theta \in [-\frac{\pi}{2}, \frac{\pi}{2}] \\ \implies \theta &= \frac{\pi}{3} \\ \implies \arcsin(\frac{\sqrt{3}}{2}) &= \boxed{\frac{\pi}{3}} \end{aligned}$$

(n) $\arccos^{-1}(-\frac{\sqrt{2}}{2})$

Solution:

$$\begin{aligned} \text{let } \arccos^{-1}(-\frac{\sqrt{2}}{2}), \text{ then} \\ \cos(\theta) &= -\frac{\sqrt{2}}{2} \text{ with } \theta \in [0, \pi] \\ \implies \theta &= \frac{3\pi}{4} \\ \implies \arccos^{-1}(-\frac{\sqrt{2}}{2}) &= \boxed{\frac{3\pi}{4}} \end{aligned}$$

2. Find the exact value of each expression.

(a) $\csc^{-1}(\cos(0))$

Solution:

$$\cos(0) = 1$$

$$\Rightarrow \csc^{-1}(\cos(0)) = \csc^{-1}(1)$$

Let $\theta = \csc^{-1}(1)$, then

$$\csc(\theta) = 1 \text{ with } \theta \in (0, -\frac{\pi}{2}] \cup (\pi, \frac{3\pi}{2}]$$

$$\Rightarrow \theta = \frac{\pi}{2}$$

$$\Rightarrow \csc^{-1}(1) = \frac{\pi}{2}$$

$$\Rightarrow \csc^{-1}(\cos(0)) = \boxed{\frac{\pi}{2}}$$

(d) $\sin^{-1}(\sec(0))$

Solution:

$$\sec(0) = 1$$

$$\Rightarrow \sin^{-1}(\sec(0)) = \sin^{-1}(1)$$

Let $\theta = \sin^{-1}(1)$, then

$$\sin(\theta) = 1 \text{ with } \theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$$

$$\Rightarrow \theta = \frac{\pi}{2}$$

$$\Rightarrow \sin^{-1}(1) = \frac{\pi}{2}$$

$$\Rightarrow \sin^{-1}(\sec(0)) = \boxed{\frac{\pi}{2}}$$

(b) $\cos \sec^{-1}(\frac{2\sqrt{3}}{3})$

Solution:

$$\text{Let } \theta = \sec^{-1}(\frac{2\sqrt{3}}{3})$$

$$\Rightarrow \sec(\theta) = \frac{2\sqrt{3}}{3} \text{ with } \theta \in [0, \frac{\pi}{2}] \cup [\pi, \frac{3\pi}{2}]$$

$$\Rightarrow \cos(\theta) = \frac{3}{2\sqrt{3}} \text{ with } \theta \in [0, \frac{\pi}{2}] \cup [\pi, \frac{3\pi}{2}]$$

$$\Rightarrow \theta = \frac{\pi}{6}$$

$$\Rightarrow \cos \sec^{-1}(\frac{2\sqrt{3}}{3}) = \cos(\frac{\pi}{6})$$

$$\Rightarrow \cos \sec^{-1}(\frac{2\sqrt{3}}{3}) = \boxed{\frac{\sqrt{3}}{2}}$$

(e) $\tan^{-1}(\sec(\pi))$

Solution:

$$\sec(\pi) = -1$$

$$\Rightarrow \tan^{-1}(\sec(\pi)) = \tan^{-1}(-1)$$

Let $\theta = \tan^{-1}(-1)$, then

$$\tan(\theta) = -1 \text{ with } \theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$$

$$\Rightarrow \theta = -\frac{\pi}{4}$$

$$\Rightarrow \tan^{-1}(-1) = -\frac{\pi}{4}$$

$$\Rightarrow \tan^{-1}(\sec(\pi)) = \boxed{-\frac{\pi}{4}}$$

(c) $\csc \sec^{-1}(\frac{\sqrt{6}}{2})$

Solution:

$$\text{Let } \theta = \sec^{-1}(\frac{\sqrt{6}}{2})$$

$$\Rightarrow \sec(\theta) = \frac{\sqrt{6}}{2} \text{ with } \theta \in [0, \frac{\pi}{2}] \cup [\pi, \frac{3\pi}{2}]$$

$$\Rightarrow \cos(\theta) = \frac{2}{\sqrt{6}} \text{ with } \theta \in [0, \frac{\pi}{2}] \cup [\pi, \frac{3\pi}{2}]$$

$$\Rightarrow \sin(\theta) = \sqrt{1 - \cos^2(\theta)}$$

$$\Rightarrow \sin(\theta) = \sqrt{\frac{1}{3}}$$

$$\Rightarrow \csc(\theta) = \frac{1}{\sin(\theta)} = \sqrt{3}$$

$$\Rightarrow \csc \sec^{-1}(\frac{\sqrt{6}}{2}) = \boxed{\sqrt{3}}$$

(f) $\sec \cot^{-1}(\frac{2\sqrt{21}}{21})$

Solution:

$$\text{Let } \theta = \cot^{-1}(\frac{2\sqrt{21}}{21}) \text{ with } \theta \in (0, \pi)$$

$$\Rightarrow \cot(\theta) = \frac{2\sqrt{21}}{21}$$

$$\Rightarrow \sec(\theta) = \frac{\sqrt{\cot^2(\theta)+1}}{\cot(\theta)}$$

$$\Rightarrow \sec(\theta) = \frac{\frac{5}{\sqrt{21}}}{\frac{2\sqrt{21}}{21}}$$

$$\Rightarrow \sec(\theta) = \frac{5}{2}$$

$$\Rightarrow \sec \cot^{-1}(\frac{2\sqrt{21}}{21}) = \boxed{\frac{5}{2}}$$

Note: using triangle to get $\sec(\theta)$ from $\cot(\theta)$ will be a good way to solve this problem.